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Project 1 – Stellar Population Synthesis Modeling of Galaxy Spectra with BC03

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1. INTRODUCTION

We present an exploratory study of stellar population synthesis modeling by fitting single stellar population (SSP) spectra from G. Bruzual & S. Charlot (2003) to empirical galaxy templates from A. L. Kinney et al. (1996) (hereafter BC03 and KC96, respectively). The goal is to investigate how well simple SSP models reproduce the spectra three distinct galaxy types: an elliptical galaxy, an Sc spiral galaxy, and a dusty starburst galaxy.

We perform fits for each galaxy type under three modeling assumptions: (1) solar metallicity without dust, (2) variable metallicity without dust, and (3) solar metallicity with dust attenuation. By comparing the resulting best fits, we examine the physical degeneracies that are present in galactic spectral modeling, particularly the age-metallicity and age-dust degeneracies.[2]

2. DATA PROCESSING

To minimize complications due to ultraviolet (UV) modeling uncertainties, we restrict our analysis to wavelengths, $3600 \leq \lambda \leq 7000$ Å, which contains prominent stellar absorption features and the 4000 Å break while avoiding the far-UV regime.

Because our goal is to model the stellar continuum, we also mask strong nebular emission lines (Table1). All wavelengths within 10 Å of each line are excluded; for the dusty starburst template, the mask is extended to 20 Å to account for stronger emission features.

To compare models and templates directly, each BC03 SSP is scaled by a normalization factor

$$a = \sum_{i=1}^{N} \frac{F_{\mathrm{KC96},i} F_{\mathrm{BC03},i}}{F_{\mathrm{BC03},i}^2}. \tag{1}$$

Goodness of fit is evaluated using the mean squared error (MSE):

$$\mathrm{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(F_{\mathrm{BC03},i} - F_{\mathrm{KC96},i}\right)^2. \tag{2}$$

In Section 3.3, we consider the effects of dust attenuation. We evaluate the goodness of fit for varying $E(B-V)$ for reddened BC03 SSPs by applying the attenuation law given in Equation 4 of D. Calzetti et al. (2000):

$$k(\lambda) = \begin{cases} 2.659\left(-2.156 + \frac{1.509}{\lambda} - \frac{0.198}{\lambda^2} + \frac{0.011}{\lambda^3}\right) + R_V & 1200 \leq \lambda \leq 6300 \text{ Å}, \\ 2.659\left(-1.857 + \frac{1.040}{\lambda}\right) + R_V & 6300 \leq \lambda \leq 22000 \text{ Å}, \end{cases} \tag{3}$$

Where we assume a constant $R_V = 3.1$. We then redden each BC03 SSP by using D. Calzetti et al. (2000) Equation 2:

$$F_{\mathrm{BC03,\,red}} = \frac{F_{\mathrm{BC03}}}{10^{0.4 E_s (B-V) k(\lambda)}}, \tag{4}$$

Where $E_s(B-V) = 0.44E(B-V)$, with $E(B-V) \in (0, 0.7)$ as our free parameter in fitting.

3. RESULTS

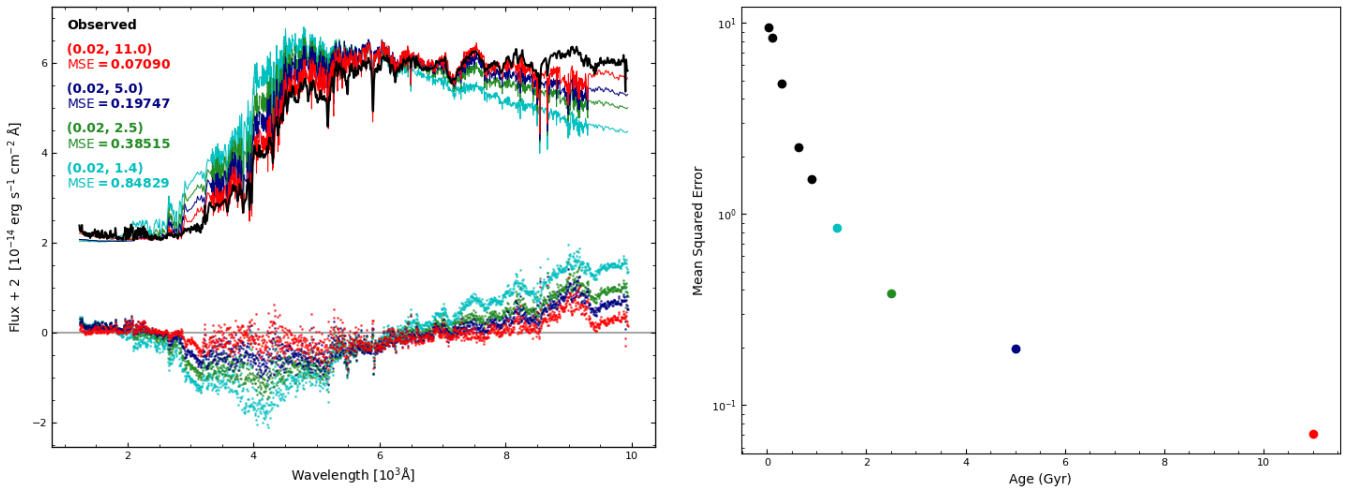
3.1. *Solar Metallicity, No Dust*

In this regime, age is the only free parameter.

[2] All code used for this project is available at https://github.com/eastcale/BC03_KC96_fitting.git

Table 1. Nebular Emission Lines

Emission Line	Wavelength (Å)
[O II]	3727
[Ne III]	2869
Hδ	4102
Hγ	4340
Hβ	4861
[O III]	4959
[O III]	5007
He I	5876
[O I]	6300
[N II]	6548
Hα	6563
[N II]	6583
[S II]	6716
[S II]	6731



(a) Spectral Fits

(b) MSE vs. Age

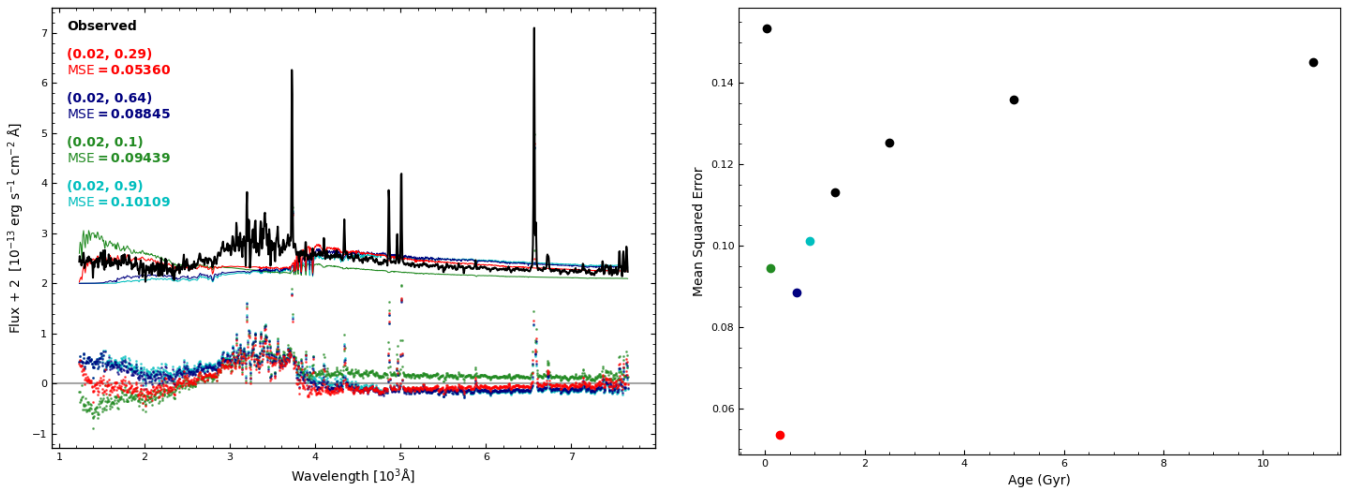
Figure 1. (a) The top four (in order: red, navy, green, cyan) best spectral fits to the KC96 elliptical galaxy template assuming no dust and solar metallicity; the residuals for each fit are shown as points in the corresponding color scattered around zero. Labels give (Z, Age [Gyr]) in the top line, and the MSE in the second line for each model. (b) MSE versus age for each of the BC03 SSPs. The top four fits are shown in the same colors as defined in (a).

3.1.1. *Elliptical*

The best-fit SSP for the elliptical template (Fig. 1) is an old population (11 Gyr), and younger models systematically produce larger MSE values. The preference for old ages is driven by the strong 4000 Å break and the red continuum slope, which are both prominent characteristics of evolved stellar populations. This result is consistent with the standard interpretation of elliptical galaxies as passively evolving systems dominated by older stars

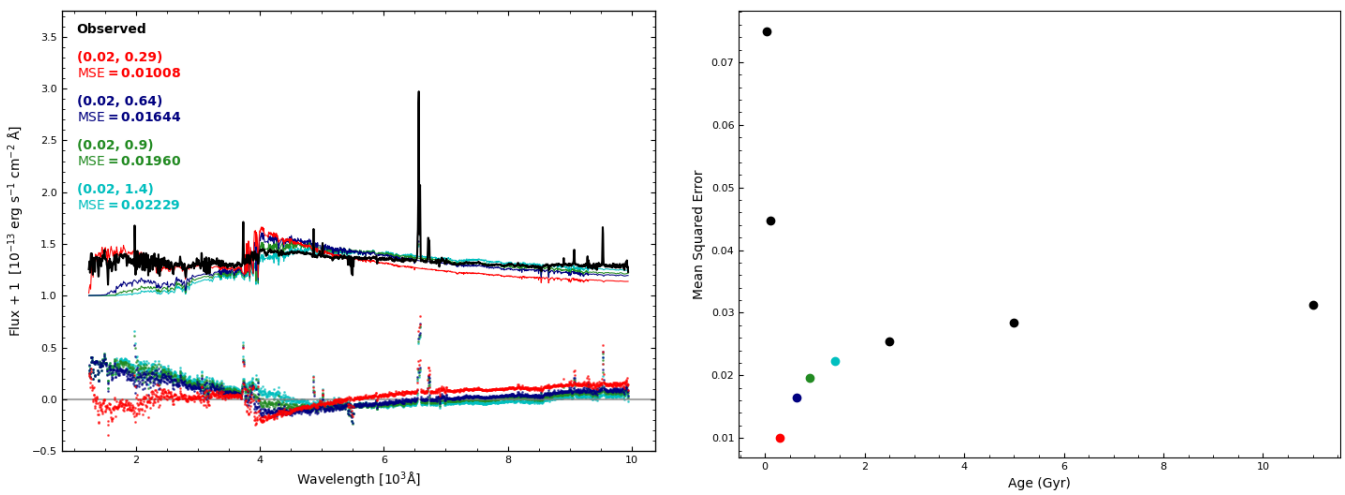
3.1.2. *Sc Spiral*

The Sc spiral template is best fit by a young population (0.29 Gyr – Figure 2). The relatively blue continuum and presence of Balmer absorption features favors younger SSPs, while older models progressively worsen the fit. However,



(a) Spectral Fits (b) MSE vs. Age

Figure 2. The same as Figure 1, but for the Sc spiral template.



(a) Spectral Fits (b) MSE vs. Age

Figure 3. The same as Figure 1, but for the dusty starburst template.

a single SSP is a simplified representation, while a true galaxy likely contains a mixture of ages. The young best-fit age should then be interpreted as a reflection of recent star formation rather than the full star formation history.

3.1.3. *Dusty Starburst*

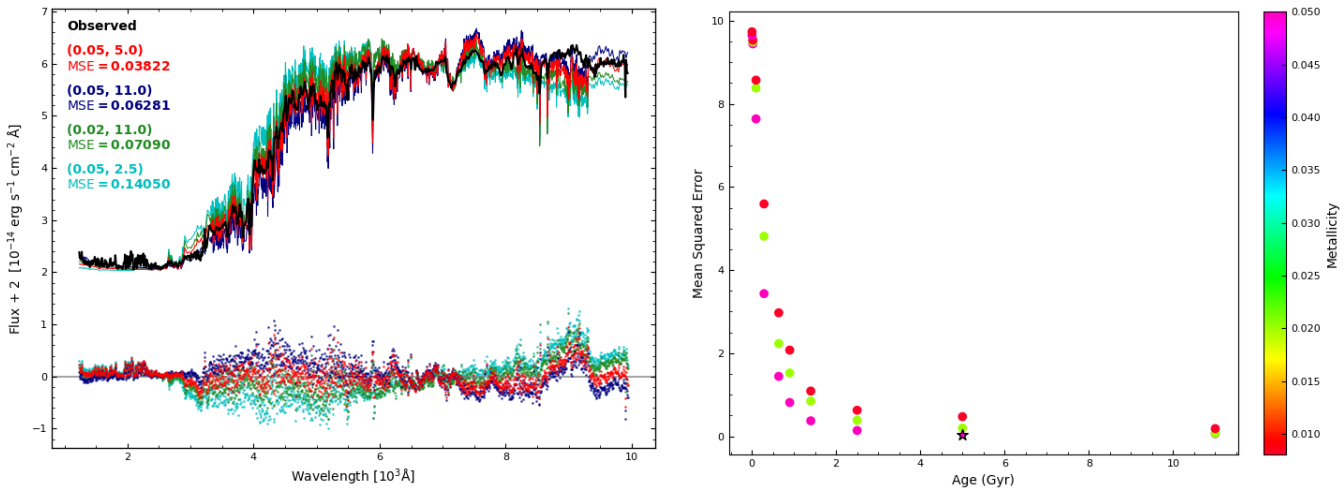
The starburst template is also best fit by a young population (0.29 Gyr – Figure 3). This is consistent with the current understanding that starburst galaxies are dominated by massive, short-lived stars that strongly influence the spectral properties. Older models provide visibly poorer matches to the continuum.

3.2. *Varying Z, No Dust*

In this regime, we allow both age and metallicity to vary.

3.2.1. *Elliptical*

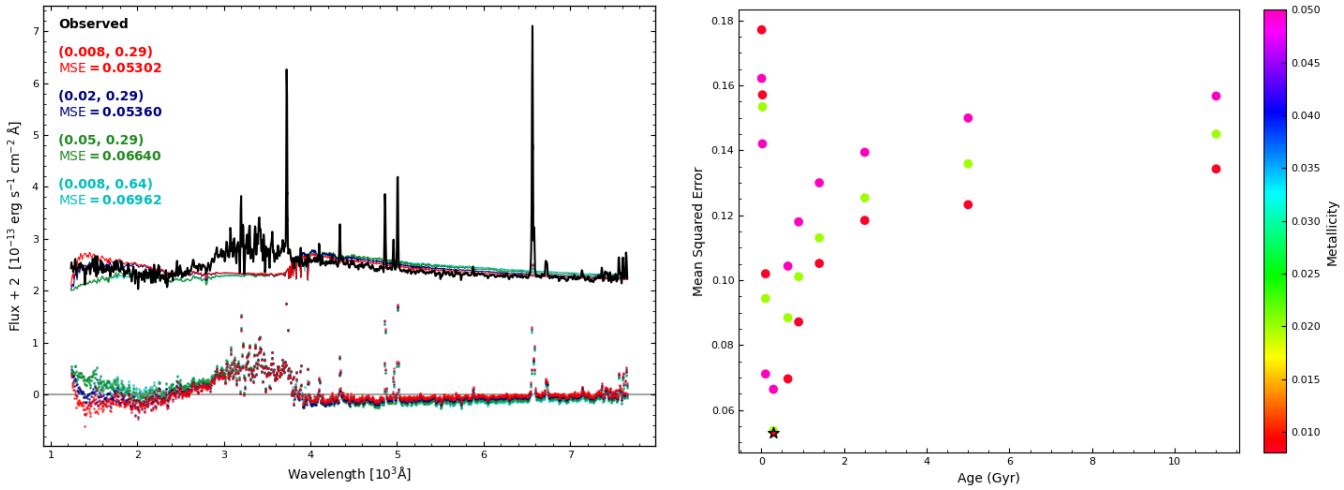
Allowing metallicity to vary reduces the best-fit age to 5 Gyr while increasing the metallicity to $Z = 0.05$ (Figure 4). This shift demonstrates the classical age-metallicity degeneracy: increasing metallicity reddens the spectrum in a



(a) Spectral Fits

(b) MSE vs. Age

Figure 4. (a) The top four (in order: red, navy, green, cyan) best spectral fits to the KC96 elliptical galaxy template assuming no dust but varying both age and metallicity; the residuals for each fit are shown as points in the corresponding color scattered around zero. Labels give (Z, Age [Gyr]) in the top line, and the MSE in the second line for each model. (b) MSE versus age for each of the BC03 SSPs. The color bar corresponds to the SSP metallicity, and the best fit is shown by the star with a black border.



(a) Spectral Fits

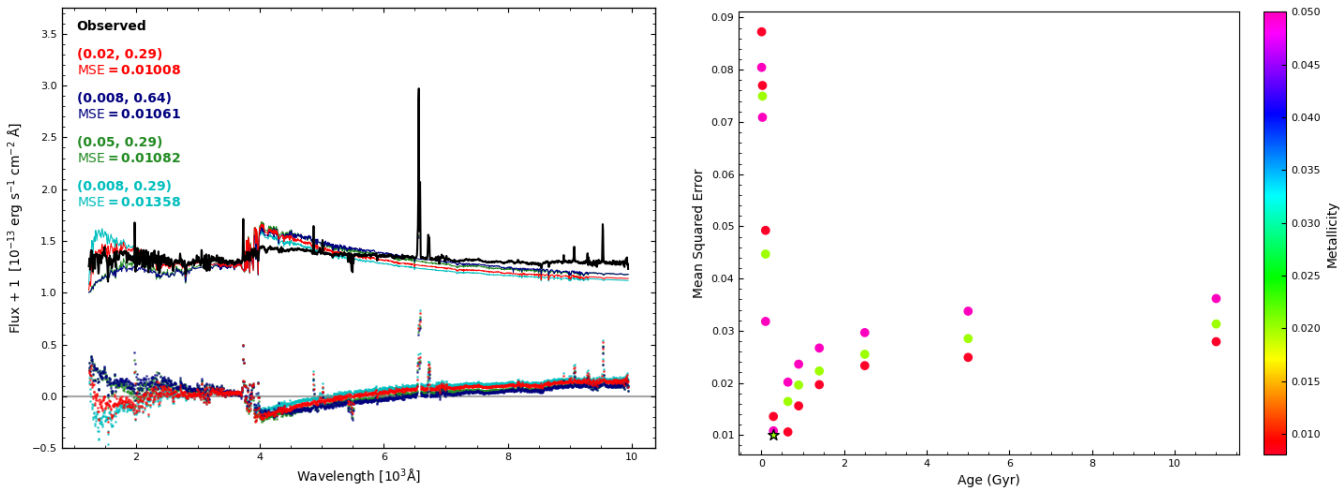
(b) MSE vs. Age

Figure 5. The same as Figure 4, but for the Sc spiral template.

manner similar to increasing age. The model compensates for a younger population by adopting a higher metallicity, producing a comparable continuum shape. The degeneracy is seen further in Figure 4b, where both young, high metallicity and old, low metallicity fits are low-MSE.

3.2.2. *Sc Spiral*

For the Sc spiral, the best fit remains young (0.29 Gyr) but shifts to a sub-solar metallicity ($Z = 0.008$) (Figure 5). The relatively blue spectrum is better reproduced by lower-metallicity models, which produce hotter stellar populations at a fixed age. The degeneracy between age and metallicity is less severe here because the stronger blue continuum tightly constrains the age. Figure 5b reinforces this, showing that the young, low metallicity fit is strongest.



(a) Spectral Fits

(b) MSE vs. Age

Figure 6. The same as Figure 4, but for the dusty starburst template.

3.2.3. *Dusty Starburst*

The starburst template remains best fit by a young solar-metallicity model (Figure 6). In this case, the spectral shape is sufficiently dominated by young stellar light that varying metallicity does not substantially alter the preferred solution.

3.3. *Solar Z, Dust Attenuation*

Here we fix $Z = 0.02$ and allow both age and reddening $E(B-V)$ to vary.

3.3.1. *Elliptical*

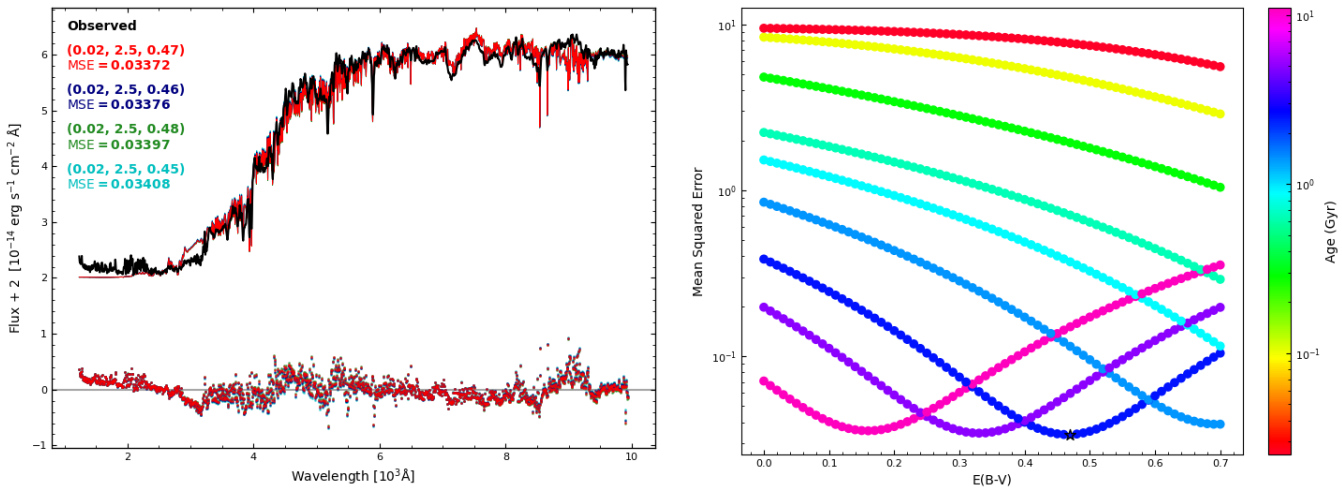
Introducing dust dramatically reduces the best-fit age to 2.5 Gyr with $E(B-V) = 0.47$ (Figure 7). This strongly reflects the age-dust degeneracy: dust reddening can mimic the continuum slope of an older stellar population. However, such a large reddening value is atypical for elliptical galaxies as they are thought to be largely without dust, suggesting that the dust-driven solution is likely over-compensating for model limitations rather than representing a physically realistic scenario. The age-dust degeneracy is very visible in Figure 7b, showing multiple low-MSE, increasing $E(B-V)$ values, each with decreasing age; this demonstrates the effect of dust making young populations appear older.

3.3.2. *Sc Spiral*

For the Sc spiral, the best-fit age decreases further to 0.10 Gyr with $E(B-V) = 0.50$ (Figure 8). The model uses dust to partially counteract the extremely blue intrinsic spectrum of a very young SSP. This again illustrates how allowing dust can bias fits toward younger ages, as reinforced by Figure 8b. Note that we only allow our $E(B-V)$ values to go up to 0.7, as the highest $E(B-V)$ presented in D. Calzetti et al. (2000) is 0.65 (we arbitrarily add 0.05 to allow extra exploration). Figure 8b shows that a higher $E(B-V)$ solution may exist, but we believe that would be less physically motivated than the fit provided.

3.3.3. *Dusty Starburst*

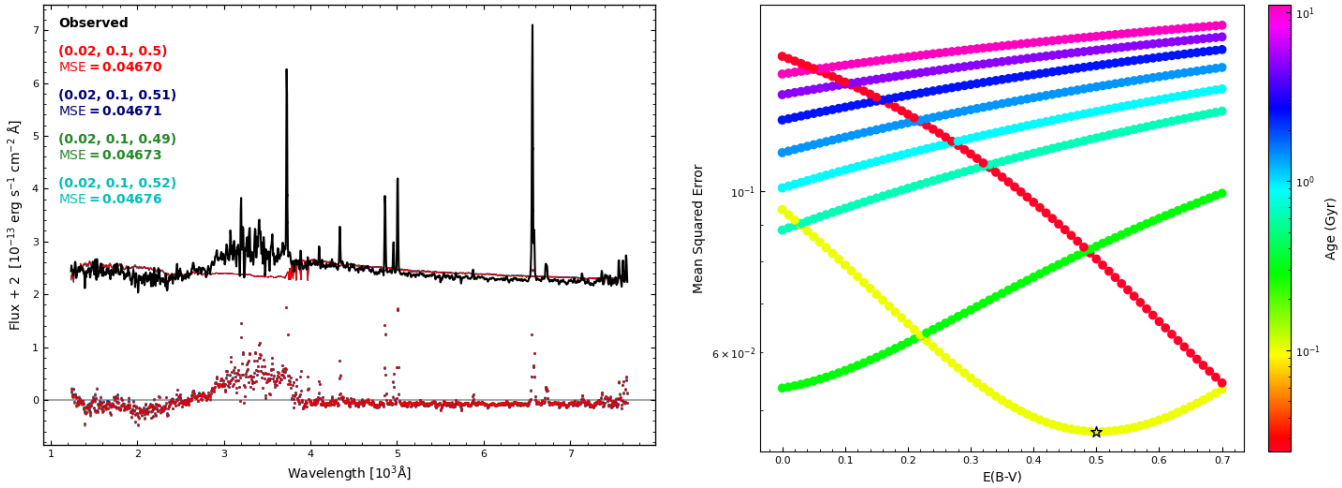
The dusty starburst is (aptly) best fit by a 0.10 Gyr population with $E(B-V) = 0.69$ (Figure 9). Unlike the elliptical case, significant dust is physically plausible in starburst systems, where intense star formation is often associated with substantial interstellar dust. The combined young age and high reddening are therefore consistent with expectations for this galaxy type. We again note the limitations of our $E(B-V)$ range on fitting in Figure 9b, but choose to keep the limit for the same reasons as discussed for the Sc spiral case.



(a) Spectral Fits

(b) MSE vs. $E(B-V)$

Figure 7. (a) The top four (in order: red, navy, green, cyan) best spectral fits to the KC96 elliptical galaxy template assuming solar metallicity and fitting for age and $E(B-V)$; the residuals for each fit are shown as points in the corresponding color scattered around zero. Labels give (Z, Age [Gyr], $E(B-V)$) in the top line, and the MSE in the second line for each model. (b) MSE versus $E(B-V)$ for each of the BC03 SSPs. The color bar corresponds to SSP age, and the best fit is shown by the star with a black border.



(a) Spectral Fits

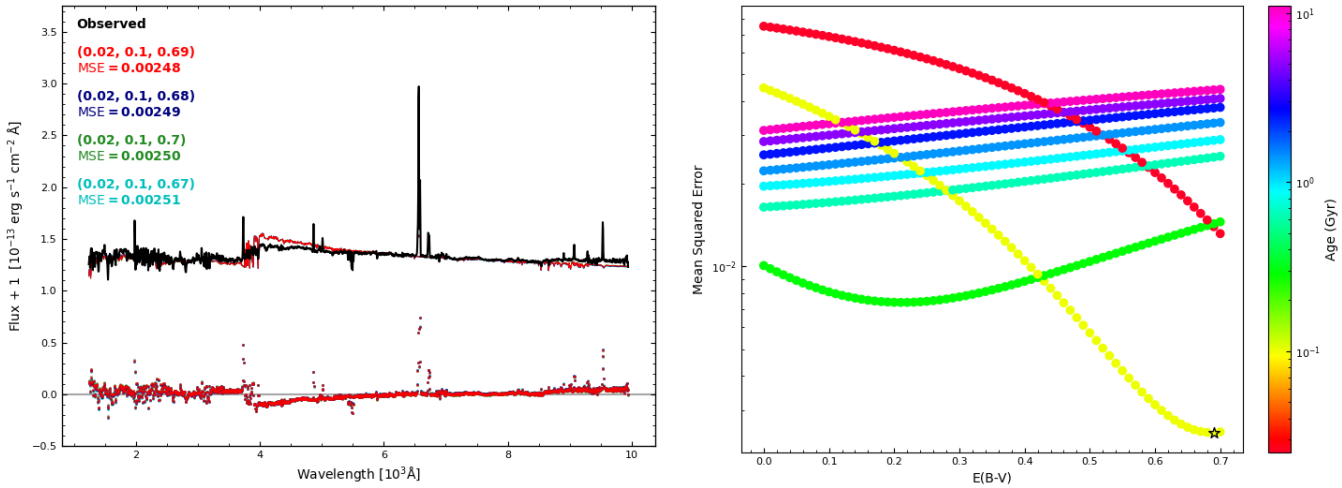
(b) MSE vs. $E(B-V)$

Figure 8. The same as Figure 7, but for the Sc spiral template.

4. SUMMARY & DISCUSSION

We fit SSP models from BC03 to galaxy templates from KC96 under three modeling assumptions. We present the results of each assumption in Table 2, where we have made bold the model we believe to be the most physically motivated for each galaxy type. Our results highlight several important degeneracies in spectral modeling of galaxies:

- **Age-Metallicity Degeneracy:** Higher metallicity can mimic older stellar populations, as seen best in the elliptical galaxy template when metallicity is allowed to vary (Section 3.2.1).
- **Age-Dust Degeneracy** Dust reddening can reproduce the continuum slope of older populations, dramatically reducing inferred ages when attenuation is included; this is also best demonstrated when the age and $E(B-V)$ are allowed to vary for the elliptical template (Section 3.3.1)



(a) Spectral Fits

(b) MSE vs. $E(B-V)$

Figure 9. The same as Figure 7, but for the dusty starburst template.

Table 2. Best-Fit Stellar Population Parameters

Galaxy Type	Model	Z	Age (Gyr)	$E(B-V)$	MSE
Elliptical	Solar Z	**0.02**	**11.0**	—	**0.07090**
	Free Z	0.05	5.0	—	0.03822
	Solar Z + Dust	0.02	2.5	0.47	0.03372
Sc Spiral	Solar Z	0.02	0.29	—	0.05360
	Free Z	**0.008**	**0.29**	—	**0.05302**
	Solar Z + Dust	0.02	0.10	0.50	0.04670
Starburst	Solar Z	0.02	0.29	—	0.01008
	Free Z	0.02	0.29	—	0.01008
	Solar Z + Dust	**0.02**	**0.10**	**0.69**	**0.00248**

- **SSP Limitations**: Real galaxies contain complex star formation histories. A single SSP fit often reflects the dominant luminous population rather than the full stellar age distribution (as discussed in Section 3.1.2).

Overall, the elliptical galaxy is consistently associated with older or higher-metallicity solutions, while the Sc spiral and starburst galaxies favor young populations. The inclusion of dust generally shifts solutions toward younger ages, emphasizing the importance of carefully constraining attenuation when modeling galaxy spectra. The elliptical galaxy template presents the strongest degeneracy, as its smooth, red spectrum is dominated by evolved stars and lacks strong Balmer absorption features that would tightly constrain age. Instead, the models must primarily match the continuum slope, the strength of the 4000 Å break, and metal absorption features. The 4000 Å break and Balmer lines are most sensitive to stellar age, metal absorption features primarily constrain metallicity, and the overall continuum slope is strongly affected by dust reddening. Because age, metallicity, and dust all influence the red continuum and break strength in similar ways, multiple parameter combinations can reproduce the observed spectrum. For these same reasons, it is particularly difficult to accurately represent the elliptical galaxy with a single SSP. This likely reflects the composite nature of elliptical stellar populations, which may contain a dominant old population with minor intermediate-age components that cannot be captured by a single burst model.

These results demonstrate both the power and the limitations of SSP fitting and highlight the need for multi-parameter modeling when interpreting galaxy spectra.

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